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Chapter 11
Leveraging Blockchain Technology in the Value Chain and the Supply Chain:
Cases on Blockchain Applications Within Chinese Firms

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ABSTRACT

The chapter will discuss and present four Chinese companies leveraging blockchain technology within their value chain and supply chain to strengthen their business and generate sustainable growth. Two of the companies described are service providers: Hyperchain and Huawei. They offer the possibility to their clients to integrate blockchain technology in order to solve business problems and improve processes within their operations. The other two cases will focus on JD e-commerce and Shunfeng express delivery, which are involved in intense supply chain activities. The descriptions will show how blockchain allowed them to develop a better, more secure, and efficient fluidity of information, capital, and goods.

DOI: 10.4018/978-1-6684-6247-8.ch011

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INTRODUCTION

Within the value chain and supply chain, it is possible to observe different flows in both directions, upstream to downstream, and vice versa (Gao, 2018). First of all, a flow of information, all companies need to make decisions and perform their operations to create and receive data (Ren et al., 2016; Mbiatem et al., 2018). Those data represent the reality of the company`s activity and operation. They could be of different natures: technological, financial, sociological, and tangibles. In all cases, for companies to be efficient, data must be tracked, accessible and most importantly trusted. Secondly, a flow of tangibles, companies possess or create assets within their operations and those assets move from one function of the value chain to another, or from one partner of the supply chain to another. To be efficient, meaning to optimize the uses of resources available to achieve a specific goal, companies need to keep track of their assets, and similarly to information, this tracking needs to be accurate and trusted (Husna et al., 2021). All resources need to be accounted for. Finally, and by far the most important, a flow of capital (or money to be simple), all resources and assets represent a monetary value (Glaa et al., 2014). Since money is limited, it Is highly controlled. In this context, financial management becomes an important function for companies to generate sustainable growth. Within a corporation, executives have an obligation to financial transparency toward shareholders and other stakeholders.

Therefore, it is of tremendous importance to make sure of the integrity of data and accuracy when reporting the flows of information, assets and money (Taghipour, 2009; Lebosse et al., 2017). It is for these reasons that blockchain technology is adopted and integrated within those flows. Blockchain offers reliability as it helps to render a truthful representation of the situation. It also offers transparency and allows permitted stakeholders the ability to inspect records. When linked to blockchain technology, data become tamperproof (Nakamoto, 2008). Blockchain could support many codified agreements between parties usually handled by traditional means, including stock trades, contracts, inventory records, quality controls, preserving provenance, maintaining the chain of custody, etc. (Taghipour and Frayret, 2013). In this way, the technology becomes part of the routine operations of the company within its value chain and supply chain (Beck, 2018).

As one of the major economic superpowers, China is not immune from this development. It is even at the forefront. However, due to the language barrier and a large amount of information, it is hard for non-Chinese readers to get a clear picture of what is happening in China. This article will help in this direction by introducing four Chinese companies and showing how they integrated blockchain technology into their business model and operations: HYPERCHAIN, a unicorn company and blockchain service providers; HUAWEI, the well-known technology company that

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provides blockchain services to clients; Shunfeng, the main express delivery company in China; and JD Group, one of the biggest e-commerce platform in China.

CASE 1: HYPERCHAIN

Introduction

HYPERCHAIN is an international leading provider of blockchain technology solutions. It is currently the first unicorn enterprise2 in the blockchain field in China. HYPERCHAIN's profit mainly comes from providing an application for blockchain technology platforms which improve the production, operation and process efficiency of clients' enterprises. As a blockchain platform development company, HYPERCHAIN provides a relatively comprehensive solution for the existing physical industry and has been applied for many situations in banking, securities, logistics platforms, and supply chain traceability. Its client's portfolio covers different fields, such as Finance with China UnionPay, Bank of China, Agricultural Bank of China, SPD Bank, and Shanghai Stock exchange; Infrastructure with State Grid, China Southern, and Haier; Technology with Huawei, Microsoft, and Google; Government with Hangzhou Government, Chongqing Government, Shanghai Government.

Founded in 2016, HYPERCHAIN first obtained 1.5 billion yuan (Chinese currency) in financing due to the release of its blockchain underlying platform products, and its cooperation with major banks in Zhejiang Province. After completing the second round of financing in 2018 and being selected as a Chinese unicorn enterprise, it cooperated with the Bank of China to complete the application of blockchain in accounts receivable and securities assetization. It was selected in the "Forbes China's Most Innovative Enterprise" list. In 2019, a joint venture was established with Yunshang IoT. HYPERCHAIN provides blockchain technology and Yunshang IoT provides hardware products, mainly selling monitoring equipment, environmental protection equipment, and sensors that deploy blockchain technology applications for clients. After that, they released their BaaS (Blockchain as a Service) platform in 2020 and cooperated with customers such as Chongqing Municipal Government and China Xiong'an Group to expand the application field of the blockchain platform. In the same year, it launched a blockchain-integrated product, integrating hardware, blockchain basic platform, and BaaS platform to help its enterprise users quickly apply blockchain technology to their enterprise management, product traceability and other aspects.

At present, there're four core products of HYPERCHAIN (*FunChain Blockchain Platform,* n.d.). First, HYPERCHAIN 's domestic self-controllable blockchain underlying platform: This product is mainly targeted at large enterprise groups,

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government agencies and industry alliances, creating a complete set of blockchain platforms that are exclusive to clients to meet corporate users' needs including security and extended application fields. Different types of personnel within the company can join the platform and set different levels of permissions to ensure the security of corporate customers. Its biggest advantage lies in the convenience of development, which means that enterprises can choose a variety of programming languages to develop blockchain applications for their unique purposes, such as mini-programs. However, the limitation is also obvious, its costs. Only large groups can therefore put up with this blockchain integration.

Second, data sharing and secure computing platform BitXMesh: This product is a new solution for data sharing. At present, in the process of informatization, enterprises have accumulated a large amount of industry data within the organization, however, the data content cannot be shared smoothly due to privacy or security issues within the enterprises, which seriously restricts the flow of data. The BitXMesh unified platform realizes the transmission method of "data available but invisible", which greatly improves the flow of data between parties. Unlike the first product, this platform allows companies to share information more easily and at a lower cost. However, the disadvantage is that companies can't develop other functions like the first product.

Third, blockchain open service platform Feilo: This product is a platform combining blockchain and cloud services. Cloud Services can greatly reduce the company's operating costs. However, at the same time, cloud services could neglect the security of the data transferred and create vulnerabilities. Under such circumstances, the Feilo platform utilizes the characteristics of blockchain technology combined with cloud services to create a better cloud experience. For example, some small and medium-sized enterprises lacking IT infrastructure capabilities can directly develop applications on the Feilo platform, which greatly reduces the operating costs of enterprises while ensuring a high level of privacy and security.

Four, Feilo Supply Chain Platform: In terms of financing, this platform uses blockchain technology to digitalize company assets, realize credit enhancement within the supply chain, and meet corporate financing needs. In terms of the supply chain, blockchain is used to supervise the entire process providing a truthful representation of the situation. For example, information about insurance and quality inspection or transfer of ownership and waybills are securely digitalized. Below are some practical examples of HYPERCHAIN products.

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Cross-Border e-Chain (Gyro Finance, 2021) Entrepot Trade Service for Banks

Entrepot trade has always been the main component of China's foreign trade, but the following problems caused great challenges for the Agricultural Bank of China in the process of financing entrepot trade enterprises:

First, the qualification level of entrepot trade enterprises is uneven because the current entrepot trade industry is not well regulated. Second, there are difficulties in verifying the authenticity of the entrepot trade documents. Banks cannot verify the authenticity and validity of relevant transactional logistics documents through an effective information platform, and they can only pay on-site visits to customs, shipping companies, and logistics departments, to verify the relevant data, which is not only difficult but also costly and risky. Third, there are Inconsistent standards for the identification of valid documents of title. According to China's current foreign exchange management policies, customs bills of lading and warehouse receipts are real and valid documents of title, but the bank found that some entrepot could not provide valid titles during the actual handling process. Four, there are difficulties in supervision in the supply chain. In entrepot trade, multiple activities such as logistics storage, import and export, and insurance, are involved, creating abundant data flows and high process complexity, so it becomes complex to achieve widespread supervision.

In this case, where the bank doubts the authenticity of this kind of entrepot trade, HYPERCHAIN proposed the following solution: "Cross-border e-chain". The main idea of its solution is based on HYPERCHAIN's blockchain underlying platform technology aimed at monitoring in real-time the information related to the flow of goods, funds and documents between banks, freight forwarders and traders, to achieve multi-party collaborative verification of authenticity. At the same time, a real-time supervision portal was opened to the government and regulatory authorities, which further helped solve the problem of information asymmetry. The "cross-border e-chain" allowed the bank to safely and confidently realize online business, which greatly improves work efficiency. In addition, the platform has also set up modules such as blacklists and credit points to keep records of dishonesty within its entire network.

In summary, the cross-border e-chain realized the synchronous sharing and verification of data and business processes through the characteristics of blockchain shared ledger, non-tampering, and traceability, which greatly reduced the reliability cost of the authenticity of entrepot trade. For the Agricultural Bank of China, it effectively solved the problem of incorrect bank information and avoided the occurrence of illegal financing by banks in entrepot trade, enhanced the enthusiasm of the Agricultural Bank to handle more financing opportunities, and effectively

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expanded financing channels for small and medium-sized enterprises. The cross-border e-chain provided many benefits to its customers and their stakeholders. The blockchain solution allowed a more comprehensive and reliable oversight of the situation with the supply chain. It also provided support for the banking industry to expand cross-border trade financing business. It ensured the standardization of trade business processes and provided more investment opportunities. The cross-border e-chain also provided benefits to different stakeholders involved in the entrepot trade process. For partners in the supply chain, it allowed them to screen out enterprises that met the qualifications and reject those that didn't. Choosing the right partner in their supply chain allowed them to improve their financial management. For regulators and governments, the cross-border e-chain platform enabled trade supervision departments to monitor the process of trade data and prevent data tampering.

Baoding Hongan Packaging Feilo Supply Chain Platform

Hebei Baoding Hongan Packaging Co., LTD. has been focusing on watermark packaging for more than 10 years. In 2015, the company updated its equipment and reached a new level of production capacity. With excellent strength and product quality, it established a good reputation in the industry and started to cooperate with large clients such as the Coffeeshop Giant in China: Luckin Coffee.

In the process of cooperation with Luckin Coffee, Baoan Hongan Packaging Company realized that it had insufficient capital reserve due to its limited scale to manage the beginning of their cooperation. To solve this issue and have financial while waiting for their client payments in the long term, they were looking for financial support. Traditionally, only bank financing was available to them to ease the financing difficulty and the company needed to provide many credit certificates. Even worse, the bank financing process takes a long time to go through its procedure and cannot guarantee the success of cooperative financing. Therefore, they were looking for a solution that could overcome those shortcomings: time and information reliability. The solution would allow instant transparency and reliability of their financial data required during the procedure required in the lending process so that their partner banks would handle the process as quickly as possible.

HYPERCHAIN proposed to use the Feilo Supply Chain platform to monitor and certify all financial and accounting information required during the financing procedures with banks and lenders. The Feilo Supply Chain platform is based on the principle of blockchain-based application of immutable traceability, which can help the funding stakeholder to check the real record of receivables, as well as the receivable evidence, leading to faster and more convenient loaning. At the same time, for active enterprises, the platform adopts an electronic seal, big data risk control and online process to facilitate financial control, monitoring and reporting.

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For Baoan Hongan Packaging Company, the Feilo supply chain with blockchain technology provided them with an excellent financing platform to handle this financing process. Now they only needed to register enterprise information on the platform, initiate billing and financing applications, and within a few hours, they can confirm billing rights from the core enterprise to the financing and lending partners. Wenjie Ge6, the head director said, "It used to take us half a year for an account to be paid back, but now with the Feilo supply chain platform, the company can receive financing support in a few hours to go through this period. It brings us great convenience in financing." In addition, the emergence of such blockchain financing platforms provided a great guarantee for investors. Thanks to the traceable technology of blockchain, investors no longer have to worry about the failure of fund recovery.

CASE 2: HUAWEI

Introduction

Huawei is a leading global provider of information and communications technology (ICT) infrastructure and smart devices. Huawei mentions that it is committed to bringing digital technology to every person, home, and organization for a fully connected, intelligent world. Huawei has 195,000 employees and operates in over 170 countries and regions, serving more than three billion people around the world (Huawei, 2021). Within this industrial giant, there is the Cloud Business Unit which is one of the newest business units established by Huawei in 2017. Zheng Yelai, the president of Huawei Cloud BU said that the cloud business in Huawei is a "pedestal". He explained that Huawei Cloud is an important channel to sell services instead of selling products. Moreover, such related businesses that Huawei has developed can be hosted on Huawei Cloud. Like carrier business, enterprise business, and consumer business (Huang, 2018).

On March 30, 2018, Huawei Cloud BU officially announced that Huawei Cloud Blockchain Service BCS was opened to the public. Huawei Cloud blockchain service (BCS) is a product at the enterprise level, that provides infrastructure service. And it has convenient, open, flexible, and efficient features. Huawei Cloud BCS can complete blockchain deployment in 5 minutes and provide a variety of secure and efficient algorithms with performance up to2000TPS~10KTPS at the same time. It can help companies quickly deploy, manage and maintain blockchain networks. It helps companies not only break the barrier of using blockchain but also focus on the development and innovation of their own business to apply the blockchain to the business quickly.

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Huawei cloud blockchain services can be used for data applications, IoT, finance, and other applications. The specific scenarios include data transactions, identity authentication, information proof of education, food traceability, IoT device management, supply chain finance management, precise poverty alleviation, etc. The application of blockchain technology can be well combined with various industry scenarios to help build technology ecosystems and digital transformation of the industry. Below is a more detailed explanation of the application of Huawei cloud blockchain services for enterprises in different industries.

Supply Chain Finance

Supply chain finance is important for small size enterprises. However, as important as it is, this process faced several issues. First of all, data security cannot be guaranteed during the process. The traditional supply chain model requires a large amount of enterprise transaction information to be provided to financial institutions, and enterprises are worried that information will be leaked by financial institutions, resulting in business secrets being learned by competitors. Then, information accuracy cannot be verified. The existing transaction data comes usually from the ERP system of the enterprise. Although the verification of the core enterprise makes the information more credible, financial institutions are still worried about authenticity, which means that the core enterprise may join hands with upstream and downstream enterprises to commit fraud.

Furthermore, the system is complex with high costs. Enterprises need to build specialized business systems for finance to connect with several financial institutions. And, different financial institutions require to build different systems based on different standards. This makes the system development and maintenance workload very difficult to manage and resource-consuming. In addition, the business process is long. The traditional supply chain model needs multiple information interactions between financial institutions, core enterprises, and upstream and downstream enterprises. The process is long, which causes it to cost a long application time and prevents enterprises in need of financing from obtaining funds quickly.

To help businesses, Huawei Cloud proposes to build a union chain of enterprises and financial institutions based on its blockchain services. It can reduce the credit risk faced by financial institutions and protect the security of enterprise data through sharing encrypted transaction information. Enterprises place accounts receivable information on Huawei Cloud's blockchain and share data with banks. The data on the blockchain is connected to the customer's local business system through a public network or dedicated line.

This system generates several benefits. It protects enterprise information security. The business data of enterprises can be used for verification in an encrypted state

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through the ability of zero-knowledge proof. It can guarantee the information security of enterprises and dispel their doubts about data. Also, the distributed ledger can guarantee the authenticity of transactions. It can guarantee that the data on the blockchain cannot be tampered with. It ensures the authenticity of the data, which helps financial institutions avoid the phenomenon of fraudulent loans. Furthermore, the deployment is fast based on Huawei Cloud technology. The system can be deployed very quickly to help financial institutions to carry out business operations in real-time. The transaction performance of open-source Fabric is several hundred tps, but Huawei can improve it to 10,000 tps through algorithm optimization. Therefore, it meets customers' requirements for timeliness to support rapid business expansion (*[HUAWEI CLOUD] Blockchain Accounts Receivable Financing Solution,* n.d.).

Intellectual Property

Managing and protecting intellectual property is an important aspect to create value and generate growth. However, the entire process and its aftermath encompass some difficulties. First of all, the time spends on confirming the intellectual property right is long. The perfect condition for the registration of ownership for intellectual property rights is when the processes not only record accurately the original work's ownership but also record all subsequent transactions involving that work (Zhuang et al., 2021).

However, the reality is that it can take several weeks or months to confirm the intellectual property right, depending on the type of IP. In addition, most creators lack effective methods to protect their works. They are unable to effectively restrict and fight against infringers. This results in a bad cycle where creators cannot use their works to create value, which in turn makes it difficult to monetize their work. It is also challenging to define infringement. To do so, it is necessary to check the authorization statement step by step to finally confirm the infringement. Especially for digital content such as sound and images, it is more likely to cause disputes of ownership.

Blockchain technology can be a solution to the issues faced by the intellectual property industry. Jiang He, a manager from Beijing Hanbang Technology Corp., said. "If the blockchain is applied in copyright protection, the protection level of intellectual property will raise." Blockchain can confirm intellectual property rights instantly through the use of blockchain certificates to create an intellectual property repository based on distributed ledger technology. When the creator uploads the original work and its details to the blockchain ledger, such information can form conclusive evidence of ownership and a timestamp.

In addition, blockchain permits efficient and cheaper management of intellectual property rights. On the one hand, blockchain technology can enable demanders and

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rights holders to establish direct peer-to-peer communication and reduce intermediate involvement. On the other hand, it can speed up the flow rate of matching supply and demand combined with big data, which greatly reduces intermediate costs and improves the efficiency of monetization. The execution efficiency of copyright charges and transactions can be accelerated through smart contracts. It can be convenient to realize payments in time by using the digital cryptocurrency attached to the blockchain system.

Furthermore, it can safeguard rights. The infringement records will be saved immutably because of blockchain technology. The infringement can be confirmed for the first time, and it can quickly find the infringing subject without the participation of inefficient third parties to arbitrate, which solves the problem of tracing. In addition, the infringement action is recorded on the blockchain which can affect the credit of the infringer and influence his transactions in the future.

Huawei has taken action in developing an intellectual property process using blockchain technology. According to a recent patent application issued by the State Intellectual Property Office, Huawei submitted a patent application for an invention in August 2016. The invention is blockchain technology that is applied to protect digital intellectual property. The system can store verification information for digital data through the blockchain. When parties initiate a download request through the peer-to-peer network, the system sends their private key and license used to access the content to match the verification information. The content will only be allowed to be downloaded after the verification request has reached a consensus on the blockchain, which means the user's private key, license, and verification information are corresponding. Huawei stated the system is a solution to protect digital content distributed through peer-to-peer networks (Ren, 2018).

On September 9, 2020, the signing ceremony of cooperation between the China Copyright Protection Center and Huawei Consumer Business was held in Shenzhen. The two sides formally signed a cooperation agreement. The cooperation will accelerate China's construction of independent innovation DCI (Digital Copyright Identifier) system. They jointly build a new ecology of the Internet copyright industry based on blockchain technology (World Wide Web, 2020). The authority of the platform means it can provide an authorization protection mechanism with tamperproof and traceability for a large number of digital works in various forms such as images, videos, audio, self-media content, and copyrights. Meanwhile, it provides a full life-cycle service in the environment of creation, including confirmation, deposition, display, evaluation, transaction, payment, and protection. The platform aims to serve copyright owners better by protecting their digital assets more effectively. The platform provides copyright owners convenience and safety in copyright deposition, distribution, trace, transaction, and protection to reduce

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the cost of judicialization of law enforcement in the field of copyright, which can monetize copyright and maximize the value of copyright[8].

E-government

On October 30, 2020, Huawei Cloud released a directory chain system. According to Zhang Xiaojun, the person in charge of the Huawei blockchain, the directory chain has been applied in more than 50 government offices in Beijing such as the Municipal Public Security Bureau and Municipal Taxation Bureau. Specifically, Huawei Cloud allows data to be shared via the blockchain platform and makes the exchange mechanism traceable and trusted. it also provides real-time feedback on data status and realizes data sharing and exchange at high speed across organizations. Moreover, all the access data can be recorded in the system. In addition, Huawei cloud also cooperated with the government of Serbia and other countries to explore the application of e-government overseas (Qilu One Pilot, 2020).

Smart Contract

On December 9, 2021, the Agricultural Bank of China Limited Shenzhen Branch and Huawei Technologies Co., Ltd. jointly held a digital RMB smart contract results conference in Shenzhen. The innovative application is in the leasing capital supervision field. It marked the successful implementation of the first smart contract application based on digital RMB.

The application aims to further strengthen the supervision of the housing leasing market, and promote stability and its healthy development. The smart contract is based on the Huawei blockchain. It uses a unified end-cloud integrated architecture and programming paradigm. What's more, the innovation of security sandbox supervision is used in the smart contract, which means it can detect, load, resolve situations, and execute automatically and safely. It provides strong technical support for the implementation of supervision as well as auditing. Meanwhile, this innovative application connects the smart contract platform and digital RMB core system, which realizes the integration between smart contracts and coin strings (*ABC Shenzhen Branch and Huawei released the industry's first digital RMB cloud-side smart contract application results,* 2021).

Future Applications

In the future, Huawei plans to apply blockchain technology to broader areas. Zhang Xiaojun, the head of Huawei blockchain, said that Huawei will build a blockchain platform to realize full cloud integration, industry digitization, and increase data

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pipelines. Full cloud means building infrastructure and BCS cloud services as well as promoting blockchain technology to apply it in more valuable scenarios. Industry digitization means constructing collaborative distributed ledgers in finance, government, energy, and transportation to optimize processes and improve efficiency. Big data pipeline means that it will build a self-organized and flat IoT network facing IoT and edge computing scenarios to support M2M secure communication, M2M transactions, and resource sharing.

CASE 3: JD GROUP (JINGDONG JITUAN)

Introduction

JD Group (*Company profile,* n.d.) is a typical B2C e-commerce platform. Different from Taobao's direct-entry model (its main competitor in China, the e-commerce branch of Alibaba), JD positions itself as a technology and service enterprise based on supply chain, and currently has businesses in retail, technology, logistics, health, insurance, production and development, and overseas fields. Jingdong Group mentioned that they adhere to the values of customer first, integrity, collaboration, gratitude, hard work and responsibility, and take technology-based, commitment to a more efficient and sustainable world as their mission. JD Group has several subsidiaries including JD Retail Group, JD Technology Group, JD Logistics Group and JD Health Group.

The profit model (*JD.com's business model*, 2020) of JD Group is based on several dimensions. First of all, JD focuses on optimizing the monitoring of its online store activities. JD collects data from customers about past sales, specific consumers click, labels, paths, views, residence time, and much more parameters. And then, they can determine and predict future consumption trends. This process, which allows JD to optimize the use of its resources, maximizes the results of its actions. In addition, they offer consumers cheaper and more reliable products. JD works directly with brand manufacturers rather than merchants like its competitor Taobao. JD's gross margin is maintained at about 5% of the generated sales on its platform, which provides more value to suppliers and customers. Furthermore, about 30 per cent of JD's profit comes from advertising, brand promotion, special events and other earnings. Up to now, nearly 80% of mainstream IT brands have launched direct cooperation with JD.

Through integrated supply chain management, JD Group gathers more upstream brands and suppliers and makes use of the scale effect of JD online mall to maximize the benefits of each industrial chain and supply chain, to create a strong B2C online retail ecosystem. For B2C platforms, the speed of the supply chain depends on the

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efficient operation of the warehouse and logistics system in the background. In recent years, JD has been taking logistics as the first task to develop, and constantly increasing the expansion of the warehouse logistics framework. They have established four logistics centres in North China, East China, South China and Southwest China, covering all major cities in the country. After just a few years, JD also founded distribution stations in more than 40 key cities. The self-built logistics system not only provides better services for customers but more importantly shortens the supply chain process and greatly reduces the operating cost. Besides, JD`s efficient leverage of logistic power can maintain its competitiveness in the market.

Their operation modes (*JD.com supply chain management analysis,* 2021) of the supply chain can be divided into the following types: FBP (Fulfillment by Platform Open Plan or POP) where JD Group shall be solely responsible for warehousing and delivery, and JD Logistics shall issue invoices; Logistic by POP (LBP) where commodities are not put into storage but are supplied by third-party sellers and delivered and invoiced by JD; SOPL (Sales on POP Logistic) where the goods are not directly stored, but the merchants deliver to the sorting centre of JD, and the merchants' issue invoices; SOP (Sales on POP) where merchants directly deliver to consumers and issue invoices.

JD has a large number of suppliers among well-known brands in china, including MUJI, Xiaomi, Huawei, Intel, etc. These enterprises directly supply goods for JD's self-run platform business and are directly delivered by JD Logistics, a subsidiary of JD Group. At present, JD has gradually formed the business model of focusing on platform self-management and supplemented it with third-party merchants.

JD has also developed strategic partnerships (WangDian, 2022). Tencent, one of the Chinese tech giants, directly injected capital into JD Group and opened its famous Chinese social media platforms such as WeChat and QQ to provide traffic entry for JD. Walmart also entered the capital and JD took advantage of Walmart's global supply chain to provide its consumers with a greater variety of imported goods. At the same time, Sam's Club entered JD. With the advantage of JD in the national supply chain, Walmart will be able to provide better logistics services for consumers. Baidu, another tech giant in China allow JD to leverage online traffic cooperation with iQiyi, a streaming video platform, to expand the influence of JD. And with Huawei, JD has formed an integrated development of hardware and software and launched a digital retail layout.

JD Zhizhen Chain Anti-Counterfeiting Traceability Service Platform (*Sina Finance, 2021*)

Through blockchain technology, JD can provide information about all sources involved in the processing of raw materials, production and circulation. Consumers

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can have access to key information by scanning code. Each link is supervised and recorded by multiple systems. Each product has a unique blockchain "ID card", and each piece of information is accompanied by a digital signature and timestamp for consumers to query and verify. Once the product is found to have a problem, the user can trace the whole chain to find the responsible subject accurately.

JD decide to establish this thorough process for several reasons (Grbrother, 2019). First of all, fake goods have always been a pain point in the e-commerce industry, and the "fake goods" problem of cross-border e-commerce has been criticized even more. JD uses blockchain technology to make goods traded across borders more "trusted." Then, people's requirements for the safety of agricultural products are constantly growing, but the sales of agricultural products sold on e-commerce platforms have not been high. So, they are not competitive compared with offline products. Furthermore, the existing purchase mode cannot complete 100% monitoring of the product quality of merchants, and product quality problems frequently occur on the JD platform. The use of blockchain technology can effectively solve the problem of product quality and accountability. Moreover, as the biggest competitor of JD, Taobao and its parent company Alibaba Group have already started the layout of blockchain anti-counterfeiting traceability technology. To combat competitors, JD Group wanted to take the lead in using the blockchain anti-counterfeiting traceability platform. JD Zhizhen Chain anti-counterfeiting traceability service platform has been used in multiple supply chains (Dr. Code, 2018) of JD Mall, including the following directions: fresh/agriculture products; Babies & Kids/wine/beauty products; 3C Used products; Luxury goods; Cross-border e-commerce; Medical products

This development by JD Zhizhen Chain's anti-counterfeiting and traceability service platform has brought several benefits. Consumers, only need to find the order on JD APP, click "One-click Tracing" or scan the tracing source code on the product directly, and then they can see the blockchain code corresponding to the transaction and know the origin of the product, the entry customs declaration number and time. For agricultural products, consumers can also intuitively understand the production and ripening process of food, so that consumers can be more assured about the quality of products purchased online

For the producer, although the use of anti-counterfeiting traceability could increase the cost of the product, the producer can accurately locate the weak points in the production process through the information record of the blockchain, to improve production efficiency. At the same time, to produce more reliable products, this platform can also improve its competitiveness in the market. For the government and supervision agencies: the launch of JD Zhizhen anti-counterfeiting traceability service platform can effectively reduce the pressure of supervision agencies on the supervision of e-commerce retail products, and assist relevant departments in

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effective supervision through transparent sharing of commodity information and authentic commodity quality.

For JD itself, blockchain has not only driven the improvement of JD's retail "trust", but also can connect all the participants in the retail supply chain, linking the key production and circulation, to achieve information transparency and tamperproof data. This helps to improve the persuasion of JD retail products and brands as well as enhance customer loyalty. Blockchain also enables JD Retail to realize more accurate sales data analysis and coordination of labour division among all participants in the supply chain, which can further promote the improvement of commodity quality, service mode and the operation efficiency of the whole supply chain. Blockchain has effectively reduced the operation cost of JD logistics. According to the official data, the application of blockchain technology in paperless logistics signing has enabled JD to save 400 million Chinese yuan in distribution labour costs throughout the year and reduce the use of paper by more than 5,000 tons.

However, at present, JD Group also encountered some difficulties (*JD.com: Blockchain technology is perfect, but there are also certain drawbacks,* 2018) in the process of developing blockchain. Liu Wenjing, head of the blockchain department, said that "the efficiency of the current blockchain is not as high as expected." At the same time, he also believes that the blockchain technology itself is not mature enough, which means that the construction, operation and maintenance of inter-subject blockchain nodes still need to reach a certain threshold. Other problems such as the efficiency of storage media needed to be solved. JD now has reached strategic cooperation with the Ministry of Industry and Information Technology, the Institute of Quality Science and Technology, the Beijing Commission of Commerce and other government agencies to join in its future development.

Even if blockchain can complete the anti-counterfeiting traceability of products, it is difficult to guarantee that the products themselves will not be replaced in the process of transportation. At present, JD hopes to achieve more strict logistics control by establishing its own logistics system. Also, the security code does not have a unified identifier. In anti-counterfeiting traceability, enterprises do not have a unified standard, and different brands of collected information are not the same. At the same time, the digital ID card could have either a one-dimensional code, two-dimensional code, RFID or other forms.

Future Layout of JD Group in Blockchain Technology *(From technology to application practice, reveal the panorama of JD.com's blockchain layout, 2020)*

JD will work on developing a BaaS platform and electronic contract. With its JD Cloud and JD Digital division, JD will promote the development of BaaS. Due to

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the high application threshold of blockchain technology, enterprises often need professional blockchain IT technicians to implement the application of blockchain. Secondly, to execute the implementation of blockchain technology, a variety of hardware resources need to be configured, and a large number of operation and maintenance personnel are needed, resulting in high application costs. The JD Zhizhen chain BaaS launched by JD Digital and JD Cloud can help customers to realize the deployment of the blockchain technology platform, and solve the practical problems encountered in the process of landing blockchain technology.

JD Zhizhen Chain Cloud is developing in the field of signed electronic contracts or smart contracts. Traditional contracts have many problems, such as high cost, slow flow and easy counterfeiting. In addition, a false seal is also a common means of counterfeiting in traditional contracts. By going electronic, costs can be cut by 80 per cent. Moreover, in terms of legal effect, an electronic contract is no different from a paper contract and is even more able to guarantee the credibility of the subject. The electronic contract architecture of JD Digital blockchain is divided into three layers: the bottom layer is blockchain service, including the JD service layer product Zhizhen chain BaaS platform; The middle tier is the depository service, which works with the authority. The top layer is the application layer, which includes initiating contracts, signing contracts, and approving flows. One advantage of JD's electronic contract is that it can be linked to the Beijing Internet Court and the Guangzhou Internet Court. When the data enters the bottom layer of the blockchain, it can be synchronized with the judicial authorities, reducing the time for collecting evidence in traditional judicial procedures. Just log in to the official website of the Internet Court and enter the certificate number to automatically synchronize the certificate evidence. Zou Zhiyou, senior manager of the information technology Department of Xiamen International Trade Group Co., Ltd. which is one of the cooperative users of JD Digital electronic contract, said that the adoption of blockchain electronic contract stems from the consideration of its characteristics such as high security, good user experience, low supply chain cost and fast efficiency improvement. The ability of JD Zhizhen Chain Cloud to sign electronic contracts to serve International Trade shares is expected to help its business process save 80% of the time cost, reducing the signing cost by 60%, which greatly improves the efficiency of trade cooperation.

CASE 4: SHUNFENG

Introduction

Shunfeng Holding Co., Ltd or SF in shorts is the largest integrated logistics service provider in China and the fourth largest in the world. The business has been

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diversified into seven segments, including time-definite express, economy express, freight, cold chain and pharmaceutical, intra-city on-demand delivery, supply chain, and international business. International business involves international express, international freight and freight forwarding, and supply chain. Revenue from time-definite express accounts for 46.4% of the total revenue. The time-definite express business provides time-definite and high-quality door-to-door delivery service for customers, enterprises, merchants, etc. it offers the options of same-day delivery, next-morning delivery, next-day delivery, etc. according to the distance and shipping route. According to the express delivery time tested by State Post Bureau in 2021, it is the fastest delivery judging by the collection-to-delivery time. Economy Express provides cost-effective and quality-guaranteed delivery services mainly for e-commerce platforms and merchants. And freight business is mainly for customers in the manufacturing and commercial distribution sectors with demand for large parcel distribution and bulk transport.

In 1993, Shunfeng was established in Shunde, Guangdong Province. It is specialized in sending express. On December 12, 2016, Shunfeng obtained approval from the Securities Regulatory Commission to be allowed to land on the A-share market, whose stock code is 002352. On February 24, 2017, it was officially renamed SF Holding Co., Ltd. SF Holding has built a brand image of "efficiency", "punctually" and "safety" through over twenty years of development. SF Holding now strives to provide a digital supply chain ecosystem leveraging technology research and development capabilities. Working with SF, there are mainly three suppliers (or major partners), the packing industry, the information technology industry, and the transportation industry. Zhejiang Dashengda Packaging Co., Ltd provides packing services for Shunfeng, Shenzhen Humeng Technology Co., Ltd provides internet information technology services, and Tianjin Shiqiao International Logistics Co., Ltd provides transportation services.

To create a smart supply chain, Shunfeng has found some important partners. For Example, Huawei, Lenovo, and Jingling group. Huawei is a leading global provider of information and communications technology (ICT) infrastructure and smart devices. Huawei provides blockchain technology to Shunfeng to help it complete the digital transformation. Lenovo is a global leader in smart devices. Lenovo provides hundreds of millions of smart devices, including computers, tablets, and smartphones, to users around the world every year. Lenovo provides traditional IT architecture such as server, storage, and network, as well as intelligent integrated architecture such as hyper-convergence, high-performance computing, SAP all-in-one, and database all-in-one for Shunfeng. Lenovo helps Shunfeng accelerate the information construction process. Jingling group is a platform in private e-commerce, which means the merchant is free from dependence on the e-commerce platform and communicates directly with the customer, and completes the commodity transaction

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through the mobile social method. Jingling group and Shunfeng will carry out comprehensive strategic cooperation in the field of e-commerce SaaS globally to promote the development of a smart supply chain. in return, Shunfeng provides solutions to the digital supply chain to Jingling group.

Also, within SF there are two major divisions involved in the development or the use of blockchain technology. First of all, Shunfeng Technology Company Limited or SF Technology, founded in 2009, is a data technology services company, which is committed to being an independent third party. SF Technology, as the intelligent brain of Shunfeng, provides the technical capabilities and knowledge to support digital transformation. Then, Shunfeng Medical Supply Chain Management Company Limited or SF Medical is a logistic company specialising in the transportation of medical supplies, which requires a high level of traceability. Both companies will be further discussed below.

Development of Blockchain Technology by SF Technology

The general direction of digital transformation of the supply chain is to shift from a producer-centric supply chain to a consumer-centric, from push to pull supply chain. It means consumers are at the basis of the supply chain. End-customers push back upstream factories and suppliers to produce and supply on demand. To achieve this efficiently, not only the supply chain should be digitally transformed but also the entire enterprise value chain processes, including finance, marketing, research, production, and management, to keep better track of the situation.

The process of connecting partners in the supply chain is already performed by SF, it is its core function since its creation. However, the traditional supply chain has several pain points, among others including how to visualize real-time logistic flows, and how to solve the exactitude and transparency of data flow between supply chain partners. These pain points are exactly what blockchain technology can solve. Shunfeng hopes to make more efficient and intelligent decisions based on the digital supply chain, instead of relying on traditional experience or other uncontrolled conditions.

To completely realize this model, it should have better data collaboration between producers and consumers, including service collaboration, data collaboration, and management collaboration. Shunfeng hopes to expand this system not only to its direct and internal activities but also to third-party partners, including third-party stores, third-party distribution partners, third-party warehouses, and third-party manufacturers. All the parties could be managed in the supply chain system together. That's one important reason why the supply chain must use blockchain to solve problems of traceability, sharing, and credibility.

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Currently, Shunfeng Technology is working on creating a supply chain collaborative management platform. The blockchain is the underlying technology in it and it solves two problems. First, blockchain ensures all the information of the supply chain can be traced and guarantee its authenticity and credibility. Second, blockchain makes the contract and document during the whole process as ledger data and assures data is shared accurately between each partner.

Shunfeng hopes to help all enterprises in the supply chain to optimize transportation and inventory. It has completed lots of algorithms and practices inside, including capacity prediction, intelligent planning, resource allocation, automatically sending vehicles, etc. In the next step, Shunfeng wishes to apply this process in more industries to collaborate and make the express industry more intelligent and efficient (*How blockchain empowers logistics development, SF Technology Logistics Conference shares practical experience,* 2021).

Use of Blockchain Technology by SF Medical for Covid-19 Vaccines

The blockchain also played an important role in cold chain transportation, especially recently for the COVID-19 vaccines. The temperature and humidity requirements are strict for the transportation of vaccines. Traditionally, the records are mostly measured and collected automatically but recorded manually. Ultimately the data is uploaded to the central database. Therefore, if there is any omission or concealment of unqualified temperature and humidity during the process, it will have a great impact on users and enterprises. The blockchain can prevent the data from being manipulated and modified.

Since 2020, SF Medical has provided transportation services for many domestic vaccine manufacturers. The Service contains vehicle transportation, vaccine quality standards, operation SOP, software for temperature routing visualization, hardware consisting of anti-theft robbery and an IoT monitoring system. The service process can be 100% monitored.

To guarantee the temperature and humidity data during the whole process, SF Technology independently developed the Baichuan supply chain chassis system and combined it with its current platform for transporting sensitive goods. The blockchain has several features including anti-tampering, anti-repudiation, and traceability. Shunfeng makes use of these characters to solidify electronic data in real-time and have efficient and safe evidence of environmental temperature and humidity data during the whole transportation. Shunfeng also cooperates with judicial institutions and authoritative institutions to dock the official notary chain to ensure the credibility of the certification.

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Chen Xiaowen, a senior technical director from SF Technology said that SF Medical, joined with SF Technology had completed all the Proof of the Existence of temperature and humidity data of COVID-19 vaccines currently. He introduced that the temperature monitoring equipment mounted on the refrigerated truck uploads temperature data every 5 minutes during the whole process. Then the Baichuan supply chain chassis system automatically uploads the whole temperature data to the Fengzheng platform finishing the Proof of Existence so that can obtain the corresponding ledger certificate of the Shenzhen Notary Office and transaction Hash for verifying uniqueness. The blockchain has secured thousands of transportation of COVID-19 vaccines since it was brought into operation in October 2021.

Chen Xiaowen pointed out that blockchain technology will integrate and optimize with other more emerging information technologies. Since blockchain technology has been put into use in various industries, it will break more and more industry barriers and reduces the difficulty of using it. For Shunfeng, they will find more application scenario and extends the existing limitations of blockchain technology to enhance the brand value.

REFERENCES

ABC Shenzhen Branch and Huawei released the industry's first digital RMB cloud-side smart contract application results. (2021, December 14). China Daily. Retrieved from http://ex.chinadaily.com.cn/exchange/partners/82/rss/channel/cn/columns/snl9a7/stories/WS61b8305ca3107be4979fcf37.html

Beck, R. (2018). Beyond bitcoin: The rise of the blockchain world. *Computer*, *51*(2), 54–58. doi:10.1109/MC.2018.1451660

Blockchain Accounts Receivable Financing Solution. (n.d.). Shangyun51. Retrieved from https://www.shangyun51.com/productdetail?id=2388

Company profile. (n.d.). JD. Retrieved from https://about.jd.com/company/

Dr. Code. (2018, October 30). *JD.com uses blockchain traceability to prevent counterfeiting? Stop tease*. Zhihu. Retrieved from https://zhuanlan.zhihu.com/p/48002623

Finance, S. (2021, June 22). *The application case of "Jingdong Zhizhen Chain Anti-counterfeiting Traceability Service Platform" was released.* Baidu. Retrieved from https://baijiahao.baidu.com/s?id=1703267168284336582

Ma Yihao (r130024174@mail.uic.edu.cn)
Downloaded: 4/24/2023 5:34:54 AM
IP Address: 218.190.235.144

From technology to application practice, reveal the panorama of JD.com's blockchain layout. (2020, November 11). Zhihu. Retrieved from https://zhuanlan.zhihu.com/p/280721643

FunChain Blockchain Platform. (n.d.). HYPERCHAIN. Retrieved from https://www.hyperchain.cn/products/hyperchain

Gao, Y., Li, Z., & Kang, D. S. (2018). Supply chain coordination: A review. *Journal of Advanced Management Science, 6*(4).

Glaa, B., Zoghlami, N., & Taghipour, A. (2014, May). Outsourcing and off-shoring healthcare services: A way to cut costs and improve quality. In *2014 International Conference on Advanced Logistics and Transport (ICALT)* (pp. 189-194). IEEE. 10.1109/ICAdLT.2014.6864116

Grbrother. (2019, June 2). *JD.com applied for 200 blockchain patents, why should JD.com develop blockchain technology?* Juejin. Retrieved from https://juejin.cn/post/6844903858771083278

Gyro Finance. (2021, June 17). *The 14th Season of Gyro Industry Blockchain | "Cross-border e-chain" entrepot trade service platform.* CSDN. Retrieved from https://blog.csdn.net/tuoluocaijing/article/details/118004638

How blockchain empowers logistics development, SF Technology Logistics Conference shares practical experience. (2021, April 14). Sina Finance. Retrieved from https://finance.sina.com.cn/tech/2021-04-14/doc-ikmxzfmk6773919.shtml

Huang, H.F. (2018). Huawei Cloud will become a cloud with Infinite Possibilities. *Communication World*, (11), 36. . doi:10.13571/j.cnki.cww.2018.11.022

Huawei. (2021). *Huawei Investment & Holding Co., Ltd. 2021 Annual Report.* https://www-file.huawei.com/minisite/media/annual_report/annual_report_2021_en.pdf?version=0401

Husna, A., Amin, S. H., & Shah, B. (2021). Demand Forecasting in Supply Chain Management Using Different Deep Learning Methods. In Demand Forecasting and Order Planning in Supply Chains and Humanitarian Logistics (pp. 140-170). IGI Global. doi:10.4018/978-1-7998-3805-0.ch005

JD.com: Blockchain technology is perfect, but there are also certain drawbacks. (2018, April 4). Sohu. Retrieved from https://www.sohu.com/a/227292954_100035093

Ma Yihao (r130024174@mail.uic.edu.cn)
Downloaded: 4/24/2023 5:34:54 AM
IP Address: 218.190.235.144

JD.com supply chain management analysis. (2021, December 3). Zhihu. Retrieved from https://zhuanlan.zhihu.com/p/440853197

JD.com's business model. (2020, April 30). Zhihu. Retrieved from https://www.zhihu.com/question/23135547

Lebosse, S., Taghipour, A., & Canel-Depitre, B. (2017). Quick response to fluctuations in supply chains: A review. *Journal of Advanced Management Science, 5*(5).

Mbiatem, J., Taghipour, A., & Canel-Depitre, B. (2018, August). Supplier selection approaches for decision makers. In *Proceedings of the 8th International Conference on Information Communication and Management* (pp. 108-112). Academic Press.

Nakamoto, S. (2008). *Bitcoin: A peer-to-peer electronic cash system.* Academic Press.

Pilot, Q. O. (2020, August 25). *HUAWEI CLOUD helps the digital city of Beijing build a catalogue blockchain.* Baidu. Retrieved from https://baijiahao.baidu.com/s?id=1675964824768948787&wfr=spider&for=pc

Ren H. (2018). Huawei applies for a blockchain technology patent. *China Plant Engineering*, (6), 6.

Ren, Z., Taghipour, A., & Canel-Depitre, B. (2016, Oct.). Information sharing in the supply chain under uncertainty. In *2016 6th International Conference on Information Communication and Management (ICICM)* (pp. 67-71). IEEE. 10.1109/INFOCOMAN.2016.7784217

Taghipour, A. (2009). Évaluation de la collaboration dans une chaîne d'approvisionnement. *Revue française de gestion industrielle, 28*(1), 26-42.

Taghipour, A., & Frayret, J. M. (2013, May). An algorithm to improve operations planning in decentralized supply chains. In *2013 International Conference on Advanced Logistics and Transport* (pp. 100-103). IEEE. 10.1109/ICAdLT.2013.6568442

WangDian, R.Z. (2022, May 19). *The history of JD.com.* Zhihu. Retrieved from https://zhuanlan.zhihu.com/p/517271496

World Wide Web. (2020, September 9). *China Copyright Protection Center and Huawei Start Cooperation to Promote Internet Copyright Ecological Governance and Innovative Development*. Baidu. Retrieved from https://baijiahao.baidu.com/s?id=1677368465142542629&wfr=spider&for=pc

Ma Yihao (r130024174@mail.uic.edu.cn)
Downloaded: 4/24/2023 5:34:54 AM
IP Address: 218.190.235.144

Zhuang, C.X., Dai, Q.Y., & Cao, J.Z. (2021). Construction of a management model for the whole life cycle of intellectual property based on blockchain technology. *China Invention & Patent*, (1), 60-65.

KEY TERMS AND DEFINITIONS

E-Commerce Platform: It refers to an internet website or a digital application where vendors and customers are connected. The business model of those platforms is about collecting commissions on every sale achieved by the vendor and by proposing digital marketing services to them to increase the probability to connect with customers.

Financial Data: It is a type of data associated with the movement of money within a company`s operations. Each financial movement or transaction will leave a trail and the characterisation of this trail represents the data.

Service Provider: It is a company or an organisation providing intangible values to customers or users in the form of a deed (actions or performances).

Sociological Data: It is a type of data associated with characteristics representing a population or a group of individuals. It could also represent the behaviours of those groups.

Supply Chain: It refers to the association of business partners joined together in sequence. Each partner provides a product or service to the next one which will in return incorporate them into their product or service. This sequence continues until the final product or service is created and delivered to the end user.

Tangible: An asset or object owned by a company that has a value. A tangible can be used, consumed, or exchanged. Those operations will leave a trail and the characterisation of this trail represents potential data.

Technological Data: It is a type of data representing the current state of a product or a machine or technological device. It could help to assess the quality level or the characteristics of a product as well as the data generated by the machines that participated in its production.

Unicorn: It is a new company that generated substantial and rapid growth. Such companies represent a huge potential for investors who invested in them since their establishment.